Abiram Srikanthasegaram

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PROFESSIONAL SUMMARY

Computer Systems Engineering undergraduate at SLIIT (2023–2027) specialising in bridging hardware and software — from register-level AVR Assembly firmware and closed-loop PID motor control, to custom PCB fabrication and fixed-wing aircraft prototyping. Proven ability to take systems from concept to tested hardware: six complete projects delivered end-to-end, each validated against measurable performance targets. Seeking an internship in Embedded Systems, Firmware Development, Control Systems, or IoT Engineering.

KEY SKILLS

Embedded Firmware AVR Assembly · C/C++ · Arduino · ATmega328P · Register-level I/O, PWM, ADC, Timer configuration · I2C · SPI · UART · STM32 (in progress)

Control Systems PID / PI control · MATLAB · Simulink · Closed-loop design · Encoder feedback · Motor control

Hardware Design PCB Design (KiCad, EasyEDA) · 3D CAD & Enclosure Design (Fusion 360) · Two-layer board fabrication · Soldering · DRC · Signal integrity

Platforms & SoC Raspberry Pi 4 · ESP32 · Arduino Uno · L298N motor driver · IR sensors & receivers

Software & Tools Python · Git · Linux · GPIO programming · Real-time GUI (Tkinter) · CSV data logging

Aeronautics Fixed-wing airframe design · Airfoil & CG analysis · Thrust-to-weight optimisation · XFLR5 · OpenVSP · NASA Glenn simulations

Electronics Digital logic · Analog circuits · BJT biasing · RC filters · Oscilloscope · Decoupling & noise filtering

EXPERIENCE

Embedded Systems Project Engineer *Nov 2023 – Present*

SLIIT – Engineering Projects | Colombo, Sri Lanka

- Accomplished stable closed-loop DC motor speed control, as measured by overshoot below 15% and steady-state error below 2%, by designing a PI controller (Kp=1, Ki=0.45) on Raspberry Pi, validating in MATLAB/Simulink, and fabricating a custom two-layer PCB.
- Accomplished fully autonomous robotic navigation with zero external libraries, as measured by 100% pass rate across 5 line-following and parking test scenarios, by programming ATmega328P entirely in AVR Assembly with a real-time FSM and IR-based parking bay detection.
- Accomplished a real-time IoT security pipeline (CLIMS), as measured by end-to-end motion-to-mobile-alert response, by integrating ESP32, PIR sensor, camera with facial recognition, electronic door lock, and push notification service.
- Accomplished full PCB lifecycle delivery, as measured by DRC-verified board and oscilloscope-validated output, by designing a 1-digit async up counter using 74HC74 flip-flops with seven-segment display in EasyEDA through to fabrication and soldering.
- Accomplished quantitative cache performance analysis, as measured by AMAT comparisons across LRU, FIFO, and Random policies at varying configurations, by building a trace-driven Python cache simulator grounded in IEEE/ACM literature.

ACADEMIC PROJECTS

PID-Controlled DC Motor Stabilisation System *Year 3, Sem 1 – 2026*

Raspberry Pi 4 · Python · L298N H-Bridge · Optical Encoder · MATLAB/Simulink · EasyEDA · Two-Layer PCB

- Designed and tuned a closed-loop PI controller (Kp=1, Ki=0.45) that brought a 12V DC motor from rest to setpoint with overshoot below 15% and steady-state error under 2% — validated in MATLAB/Simulink simulation before committing to hardware.
- Fabricated a custom two-layer PCB to replace breadboard wiring, isolating encoder signal paths from high-current motor traces — directly eliminated intermittent encoder noise and voltage-drop instability that caused controller divergence during prototyping.
- Built a real-time Python monitoring application featuring live RPM graphing, dynamic Kp/Ki adjustment sliders, PWM duty cycle display, and automated CSV logging — enabling iterative tuning without reflashing firmware.
- Made a deliberate engineering decision to omit the derivative term: experimental testing confirmed Kd amplified encoder quantisation noise without improving rise time or reducing overshoot — PI-only design was justified and documented.

Line-Following Arduino Car with Autonomous Parking *Year 3, Sem 1 – 2026*

AVR Assembly · ATmega328P · Arduino Uno · L298N · 3× IR Line Sensors · 38 kHz IR Receiver

- Programmed a fully autonomous mobile robot entirely in AVR Assembly with zero library dependencies — all peripheral configuration (DDR/PORT/PIN registers, Timer0 Fast PWM, ADC) written at register level.
- Designed an 8-state FSM (Initialisation → Line Following → Marker Detection → Slow Mode → Parking Detection → Parking Execution → Lost Line Recovery → Stop) achieving 100% pass rate across all 5 test scenarios.
- Engineered analog IR parking detection using ADC threshold comparison on channel A0, triggering a deterministic 7-step parking manoeuvre — chosen over digital detection to provide configurable sensitivity against ambient IR interference.
- Implemented lost-line recovery with alternating left/right swing searches and bounded retry counts — preventing indefinite off-track driving during sharp turns or surface glare events.

Cache Design & Performance Analysis Simulator *Year 2 – 2025*

Python · Computer Architecture · Trace-Driven Simulation · AMAT Analysis

- Built a configurable trace-driven cache simulator sweeping 36 parameter combinations — line sizes (16B–128B), associativity (1–4-way), replacement policies (LRU, FIFO, Random) — grounded in IEEE/ACM literature.
- Identified 64B as optimal cache line size: miss ratio improved from 0.237 at 16B to 0.155 at 64B (35% reduction), degrading at 128B due to over-fetch and cache pollution — confirming spatial locality sweet-spot hypothesis.
- Determined optimal L1 configuration: 64B lines, 2-way associativity, LRU policy achieving AMAT ~13.0 ns — aligning with industry-standard L1 design practice.

CLIMS – Smart Surveillance & Access Control System *Year 1 – 2024*

ESP32 · IoT · Python · OpenCV Facial Recognition · Electronic Door Lock · Mobile Push Notifications

- Designed and built a complete end-to-end IoT security pipeline: PIR motion trigger → camera capture → facial recognition → electronic lock actuation → real-time mobile push notification — full decision workflow on a single embedded platform.
- Engineered face-based access control to distinguish known from unknown faces before lock actuation — preventing false positives from ambient motion.

1-Digit Asynchronous Up Counter – PCB Design & Fabrication *Year 2 – 2025*

EasyEDA · 74HC74 D Flip-Flops · Seven-Segment Display · PCB Fabrication · Oscilloscope Validation

- Completed the full hardware lifecycle for a 0–9 asynchronous decimal counter: EasyEDA schematic → PCB layout → DRC sign-off → board fabrication → soldering → oscilloscope validation of switching waveforms at each flip-flop stage.

PERSONAL PROJECTS

Agni Wing – Fixed-Wing Aircraft Prototype *2025 – Ongoing*

Airframe Design · A2212 1125KV BLDC · 40A ESC · 3S 2200mAh LiPo · 1045/1047 Propeller · Fusion 360 CAD · NASA Glenn Simulations

- Co-founded a four-person self-directed engineering team building an autonomous fixed-wing UAV — beginning with a manually controlled prototype to validate aerodynamic fundamentals before integrating flight control electronics.
- Completed full airframe design cycle within 6 days: aerodynamic layout, fuselage sizing, CG calculation, propeller offset, thrust-line alignment, and wing-fuselage integration using NASA Glenn's Beginner's Guide to Aeronautics and Fusion 360 CAD.
- Ran 5 iterative build-test-crash-analyse-rebuild cycles — Attempt 5 achieved ~12 seconds of sustained flight at 30–40 ft altitude, validating CG positioning and thrust-to-weight ratio (AUW ~850–900 g) within flyable margins.
- Each failed attempt produced structured failure analysis: Attempts 1–3 identified CG instability and insufficient thrust; Attempt 4 revealed wing-root structural weakness; Attempt 5 incorporated all corrections and achieved first controlled flight.

CERTIFICATIONS & TRAINING

PLC & Industrial Automation Workshop (PLCFI) *2025*

IEEE Student Branch, Curtin University Colombo · Sri Lanka Institute of Robotics (SLIR)

- Completed hands-on sessions in PLC fundamentals, ladder logic programming, HMI/SCADA configuration using XINJE PLC hardware, and live motor and sensor control demonstrations. Awarded certificate of completion by SLIR.

EDUCATION

BSc (Hons) Computer Systems Engineering *Nov 2023 – Nov 2027 (Expected)*

Sri Lanka Institute of Information Technology (SLIIT)

- Relevant modules: Embedded Systems Design · Control System Engineering · Digital Electronics · Computer Architecture · Analog Electronics · Systems Modelling & Prototyping

REFERENCES

Three confirmed referees — full contact details available upon request.

Mr. Dinith Primal
Head of Department – Computer Systems Engineering
MSc Wireless Communications
Sri Lanka Institute of Information Technology (SLIIT)
Relationship: Head of Department — confirmed as referee

Mr. Sanjeeva Perera
Senior Academic Fellow – Faculty of Computing
BSc Eng (University of Moratuwa) · MBA (University of Colombo)
Sri Lanka Institute of Information Technology (SLIIT)
Relationship: IE3014 Professional Skills lecturer — confirmed as referee

Ms. Dinithi S. Pandithage
Lecturer & Year 2 CSNE Coordinator – Faculty of Computing
BSc (Hons) Computer Engineering (KDU) · MIEEE · MIET
Sri Lanka Institute of Information Technology (SLIIT), Malabe Campus
Relationship: IE3064 Embedded Systems Design lecturer — supervised AVR Assembly robot project, confirmed as referee